

```

let is-blank(val) = {
  if val == none { true }
  else if type(val) == "string" and val == "" { true }
  else { false }
}
let values = d.map(p => p.at(1))
let raw-max = calc.max(..values)
let max-val = if raw-max == 0 { 10 } else { calc.max(10, calc.ceil(raw-max / 10) 10) }
let has-title = not is-blank(title)
let has-x-label = not is-blank(x-label)
let has-y-label = not is-blank(y-label)
let svg-width = 440
let svg-height = if has-x-label { 225 } else { 200 }
let x-start = if has-y-label { 65 } else { 50 }
let x-end = 415
let plot-width = x-end - x-start
let y-bottom = 150
let y-top = 20
let plot-height = y-bottom - y-top
let num-items = d.len()
let slot-width = plot-width / num-items
let points-arr = ()
let elements-arr = ()
let i = 0
for item in d {
  let label = item.at(0)
  let val = item.at(1)
  let slot-center = x-start + (i slot-width) + (slot-width / 2)
  let x = slot-center
  let h = (val / max-val) plot-height
  let y = y-bottom - h
  points-arr.push(str(x) + "," + str(y))
  elements-arr.push("<circle cx=" + str(x) + " cy=" + str(y) + " r='4' fill=" + color + "><text x=" + str(slot-center) +
    " y=" + str(y - 8) + " font-size='10' text-anchor='middle' fill=" + color + ">" + str(val) + "</text><text x=" +
    str(slot-center) + " y=" + str(y-bottom + 20) + " font-size='11' text-anchor='middle' fill='#666'>" + label +
    "</text>")
  i += 1
}
let polyline-svg = "<polyline points=" + points-arr.join(" ") + " fill='none' stroke=" + color + " stroke-width='2.5'>"
let trend-svg = if trend-line and num-items > 1 {
  let sum-x = 0
  let sum-y = 0
  let sum-xx = 0
  let sum-xy = 0
  let idx = 0
  for item in d {

```

```

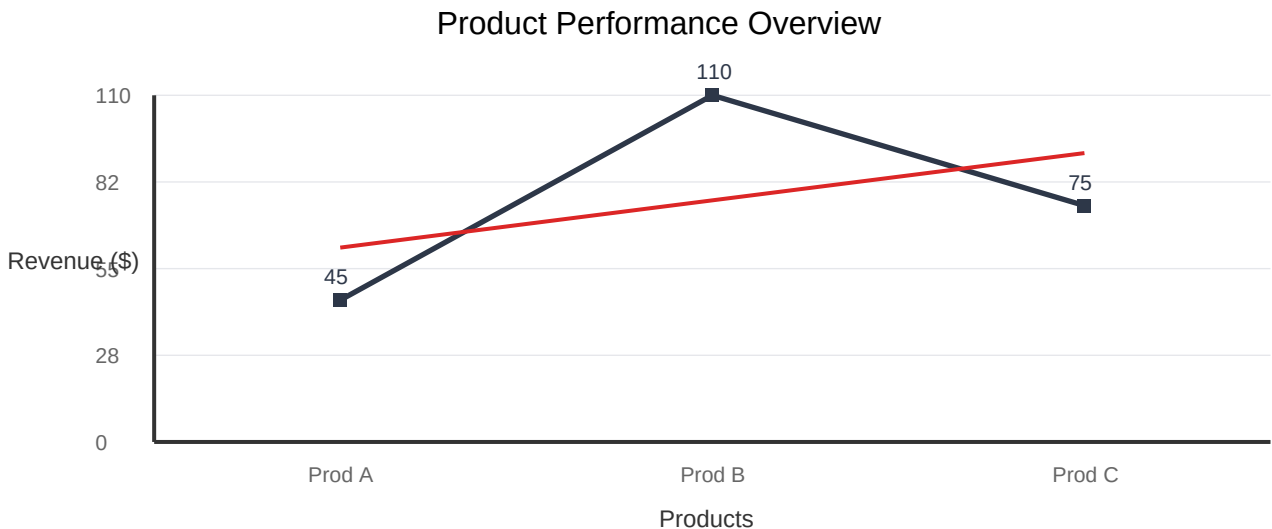
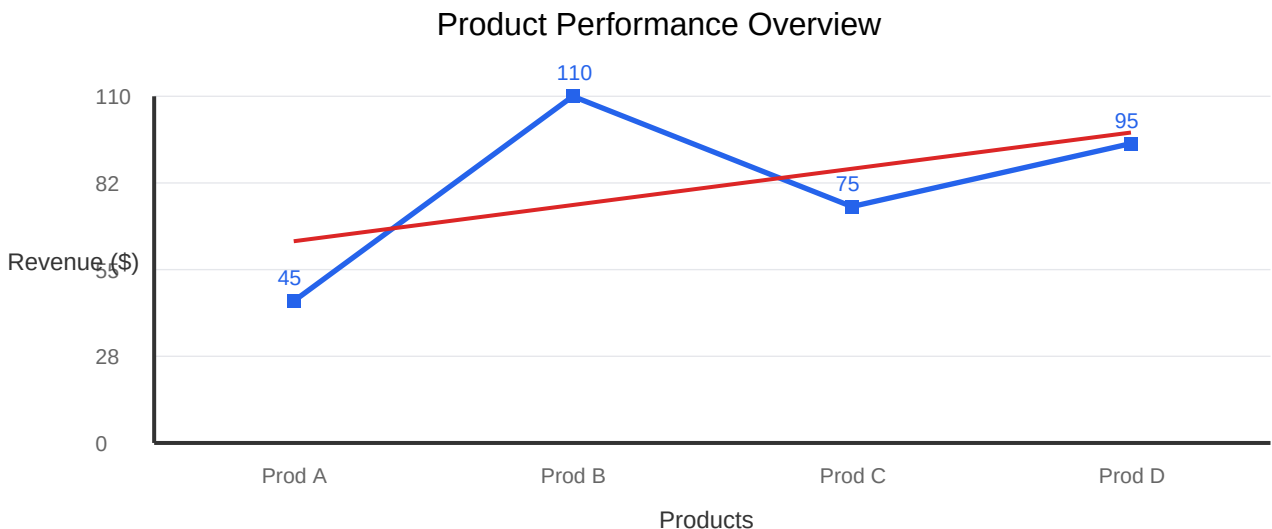
let x-val = idx
let y-val = item.at(1)
sum-x += x-val
sum-y += y-val
sum-xx += x-val x-val
sum-xy += x-val y-val
idx += 1
}
let denom = (num-items sum-xx) - (sum-x sum-x)
if denom != 0 {
let m = (num-items sum-xy - sum-x sum-y) / denom
let b = (sum-y - m sum-x) / num-items
let y1-val = m 0 + b
let x1-px = x-start + (0 slot-width) + (slot-width / 2)
let y1-px = y-bottom - ((y1-val / max-val) plot-height)
let y2-val = m (num-items - 1) + b
let x2-px = x-start + ((num-items - 1) slot-width) + (slot-width / 2)
let y2-px = y-bottom - ((y2-val / max-val) plot-height)
"<line x1='" + str(x1-px) + "' y1='" + str(y1-px) + "' x2='" + str(x2-px) + "' y2='" + str(y2-px) + "' stroke='#dc2626'
stroke-width='2' stroke-dasharray='4'/">"
} else {
""
}
} else {
""
}
}
let grid-svg = range(5).map(idx => {
let val = (max-val / 4) (4 - idx)
let y = y-top + (idx (plot-height / 4))
"<line x1='" + str(x-start) + "' y1='" + str(y) + "' x2='" + str(x-end) + "' y2='" + str(y) + "' stroke='#e5e7eb'
stroke-width='1'/"><text x='" + str(x-start - 8) + "' y='" + str(y + 4) + "' font-size='10' text-anchor='end' fill='#666'">" +
str(calc.round(val)) + "</text>"
}).join("")
let y-label-svg = if has-y-label {
let cy = y-top + (plot-height / 2)
"<text x='20' y='" + str(cy) + "' transform='rotate(-90, 20, '" + str(cy) + "')' font-size='11' text-anchor='middle'
fill='#333'">" + str(y-label) + "</text>"
} else {
""
}
let x-label-svg = if has-x-label {
let cx = x-start + (plot-width / 2)
"<text x='" + str(cx) + "' y='" + str(y-bottom + 42) + "' font-size='11' text-anchor='middle' fill='#333'">" + str(x-label)
+ "</text>"
} else {
""
}

```

```

}
let svg-string = "<svg viewBox='0 0 " + str(svg-width) + " " + str(svg-height) + "
xmlns='http://www.w3.org/2000/svg'><rect width='100%' height='100%' fill='white'/>" + grid-svg + "<line x1='" +
str(x-start) + "' y1='" + str(y-bottom) + "' x2='" + str(x-end) + "' y2='" + str(y-bottom) + "' stroke='#333'
stroke-width='2'/><line x1='" + str(x-start) + "' y1='" + str(y-top) + "' x2='" + str(x-start) + "' y2='" + str(y-bottom) + "'
stroke='#333' stroke-width='2'/>" + trend-svg + polyline-svg + elements-arr.join("") + y-label-svg + x-label-svg +
"</svg>"
let chart-content = image(bytes(svg-string), format: "svg", width: 80%)
align(center, if has-title {
stack(
spacing: 8pt,
if type(title) == "function" { title() } else { title },
chart-content
)
} else {
chart-content
})
}

```



## Prhmance Overview

